

Predicting the Compressive Strength of High-Performance Concrete

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Springboard Data Science Capstone Project

1. Introduction

High-performance concrete (HPC) is widely used in modern construction due to its superior mechanical and durability characteristics. Unlike conventional concrete, HPC incorporates additional engineered components—such as blast furnace slag, fly ash, and superplasticizers—that enhance performance but also increase the complexity of its behavior. Due to its compositional complexity, the properties of HPC are difficult to model.

This project investigates two central questions:

- Which material and mixture factors most strongly influence compressive strength?
- Can we accurately predict compressive strength from composition and age using machine learning?

To answer these, a supervised regression approach was applied to a well-known dataset of concrete mixtures and their measured strengths.

2. Data Overview

The dataset, sourced from the UCI Machine Learning Repository, originally contained 1030 entries, reduced to 996 after deduplication. Each entry represents a concrete mixture with:

Features (kg/m^3 unless noted):

- Cement
- Blast Furnace Slag
- Fly Ash
- Water
- Superplasticizer
- Coarse Aggregate
- Fine Aggregate

- Age (days)

Target Variable:

- Compressive Strength (MPa)

This is a classic supervised regression problem.

3. Feature Engineering

Prior research in concrete science highlights the importance of mixture ratios rather than raw component quantities. The project incorporated several engineered features based on known material science relationships, including:

- Water:Cement
- Water:Binder
- Superplasticizer:Binder
- Fly Ash:Binder
- Slag:Binder
- (Slag + Fly Ash):Binder

These ratios help capture the chemical and physical interactions that govern strength development.

4. Data Preprocessing

To prepare the data for modeling:

- Row-wise normalization was applied to mixture components, converting them into fractions of the total mixture.
- Age was scaled using min-max normalization.
- The dataset was split into 80% training and 20% testing subsets.

This ensured that models were not biased by differing scales among features.

5. Modeling Approach

Three regression models were selected to compare performance:

1. Linear Regression
2. Random Forest Regressor
3. Gradient Boosting Regressor

Model Evaluation

Models were tuned and evaluated using:

- 5-fold cross-validation
- R^2
- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)

Hyperparameters

Key tuned parameters included:

Random Forest

- `n_estimators = 250`
- `max_depth = None`
- `min_samples_leaf = 1`
- `min_samples_split = 2`

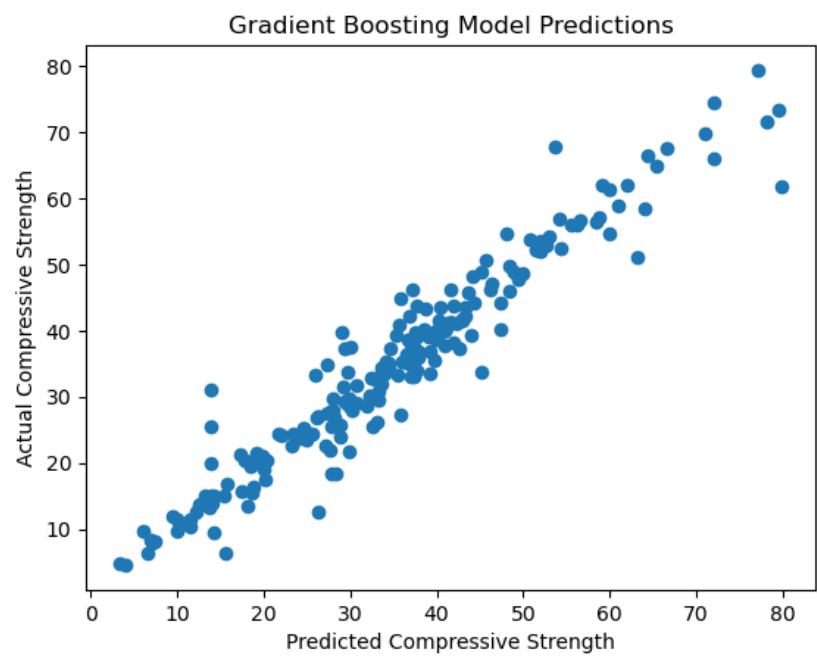
Gradient Boost

- `n_estimators = 500`
- `learning_rate = 0.2`

6. Results

Model	R ²	MAE	RMSE
Linear Regression	0.66	7.42	9.25
Random Forest	0.92	3.32	4.52
Gradient Boost	0.93	2.81	4.26

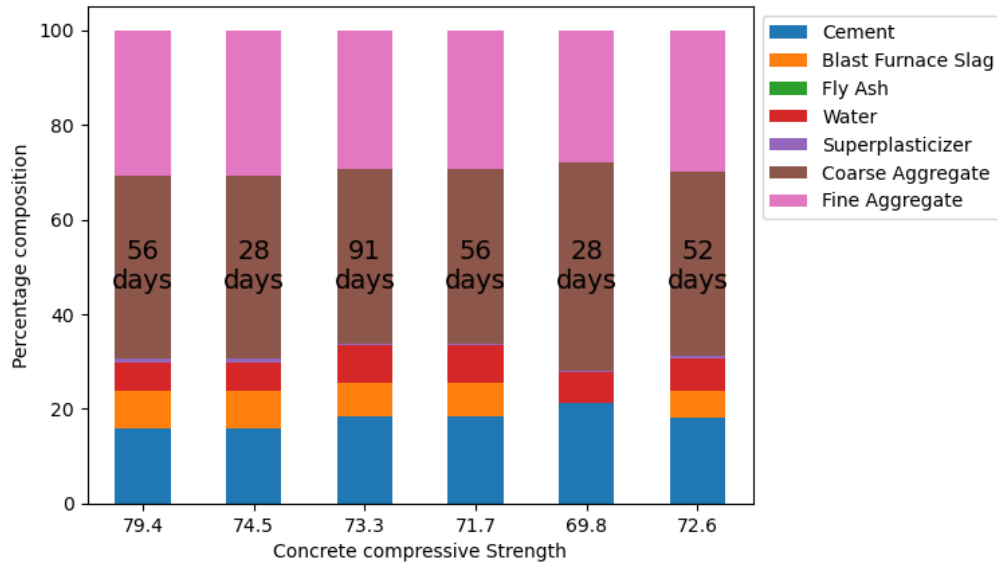
The Gradient Boosting model achieved the best overall performance, with the highest R² and lowest error metrics. This indicates that nonlinear relationships and interactions among mixture components are important for predicting strength.



7. Final Analysis

To explore optimal mixture design, the project used the scipy optimize module to search for feature combinations that maximize predicted compressive strength under the Gradient Boost model. This step demonstrates how machine learning models can support practical engineering decisions—such as designing mixtures for maximum strength—beyond simple prediction.

Top five compositions from test set compared to optimized composition (right)



8. Conclusions

This project demonstrates that machine learning can effectively model the compressive strength of high-performance concrete, even with its complex mixture design.

Key Findings

- Gradient Boosting was the most accurate model, achieving an R^2 of 0.93.
- Water-cement ratio and age were consistently identified as major predictors of strength.
- Feature engineering based on material science principles significantly improved model performance.
- Optimization techniques can be used to propose high-strength mixture designs.

Recommendations

The findings of this project offer several practical applications for clients working with high-performance concrete:

1. Optimize Mix Designs Prior to Production

The Gradient Boost model's strong predictive accuracy enables clients to evaluate alternative mixture proportions virtually. By simulating different combinations of cementitious materials, water, and admixtures, clients can identify mixtures that

maximize compressive strength while reducing material waste and laboratory testing time.

2. Enhance Quality Control Processes

The analysis highlights water-cement ratio and age as dominant predictors of strength. Clients can use these insights to establish tighter control limits, flag potentially underperforming batches earlier in the production process, and supplement traditional QC testing with model-based predictions.

3. Support Engineering and Proposal Development

Model-generated strength predictions allow engineering teams to compare design alternatives and justify material choices with quantitative evidence. This strengthens technical proposals, improves communication of trade-offs, and supports more informed decision-making in project planning.

Future Work

A more comprehensive evaluation of concrete performance would include additional properties such as:

- Tensile strength
- Durability
- Thermal resistance

Incorporating these would enable a more holistic model of HPC behavior.